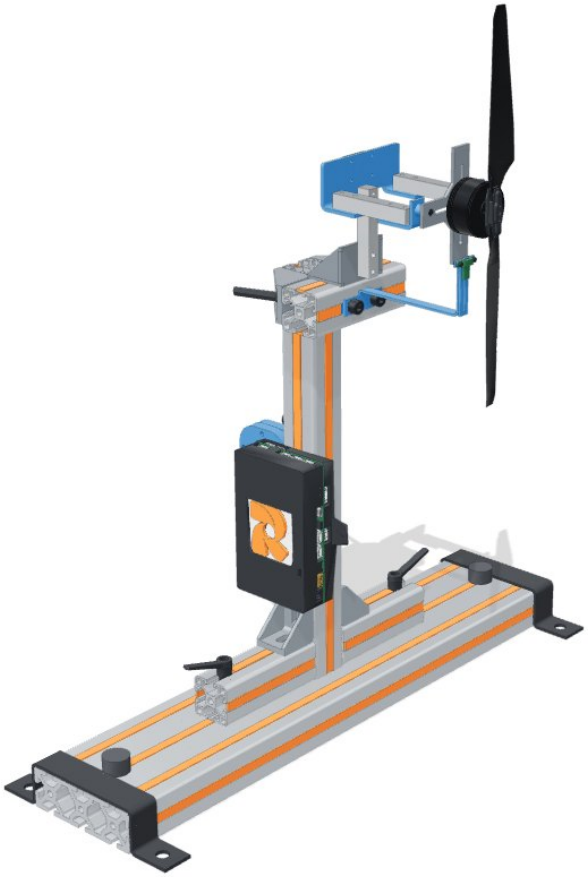
DynoX thrust stand

RTS Compact 5

User manual · 5 kgf thrust and torque test stand



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Designed and manufactured by REUDE Technologies Private Limited, Chennai, India. Read the safety instructions in Section 3 before assembling or operating this equipment.

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1. About this manual

This manual explains how to assemble, wire, calibrate, operate and maintain the RTS Compact 5. It is written for technicians, engineers and lab staff with basic mechanical and electrical skills. Keep it with the stand and pass it on with the equipment.

How to read the safety notes

WARNING A hazard that can cause serious injury. Follow the instruction exactly.

CAUTION A risk of damage to the stand, the sensors or the measurement.

NOTE Useful information that makes the job easier or the result better.

Interactive 3D model

Every component in this manual is numbered the same way as in the DynoX 3D viewer. Open the viewer to rotate the stand, see the exploded sub-assemblies, and lay every component out on the table.

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2. Product overview

The RTS Compact 5 is a bench-top thrust stand for drone motors and propellers up to 5 kgf of thrust. It measures thrust, torque, motor speed, current, voltage and vibration together, and streams the data to the ROTRIX software for live monitoring, logging and performance reports.

Specifications

Item	Specification
Rated thrust	0 to 5 kgf
Measured parameters	Thrust, torque, RPM, voltage, current and vibration
Force sensing	One thrust beam load cell and two torque beam load cells
Speed sensing	Non-contact optical RPM sensor on an adjustable arm
Electrical sensing	Inline current sensor module; voltage measured by the DAQ unit
Data acquisition	ROTRIX DAQ unit on the side of the mast (same unit as the RTS Pro 20); USB link to the PC
Software	ROTRIX test control and logging; ROTRIDASH analytics (optional)
Frame	40-series aluminium extrusions with orange slot covers (REUDE standard)
Footprint	About 600 × 205 mm (base rail with bench clamps)
Thrust axis height	About 525 mm above the bench
Reference propeller	About 340 mm (13 inch) folding propeller

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Main components

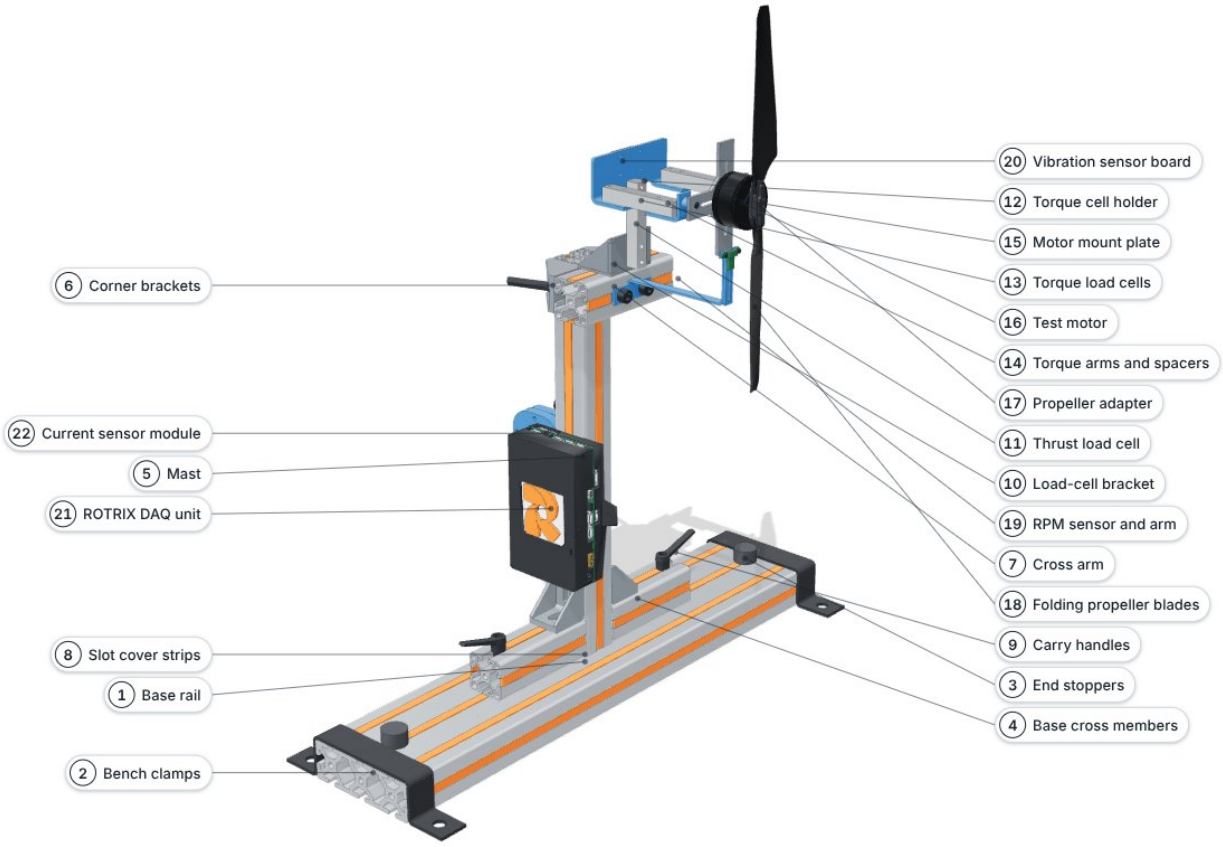


Figure 2.1 RTS Compact 5 assembled, with the main components numbered.

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Exploded view

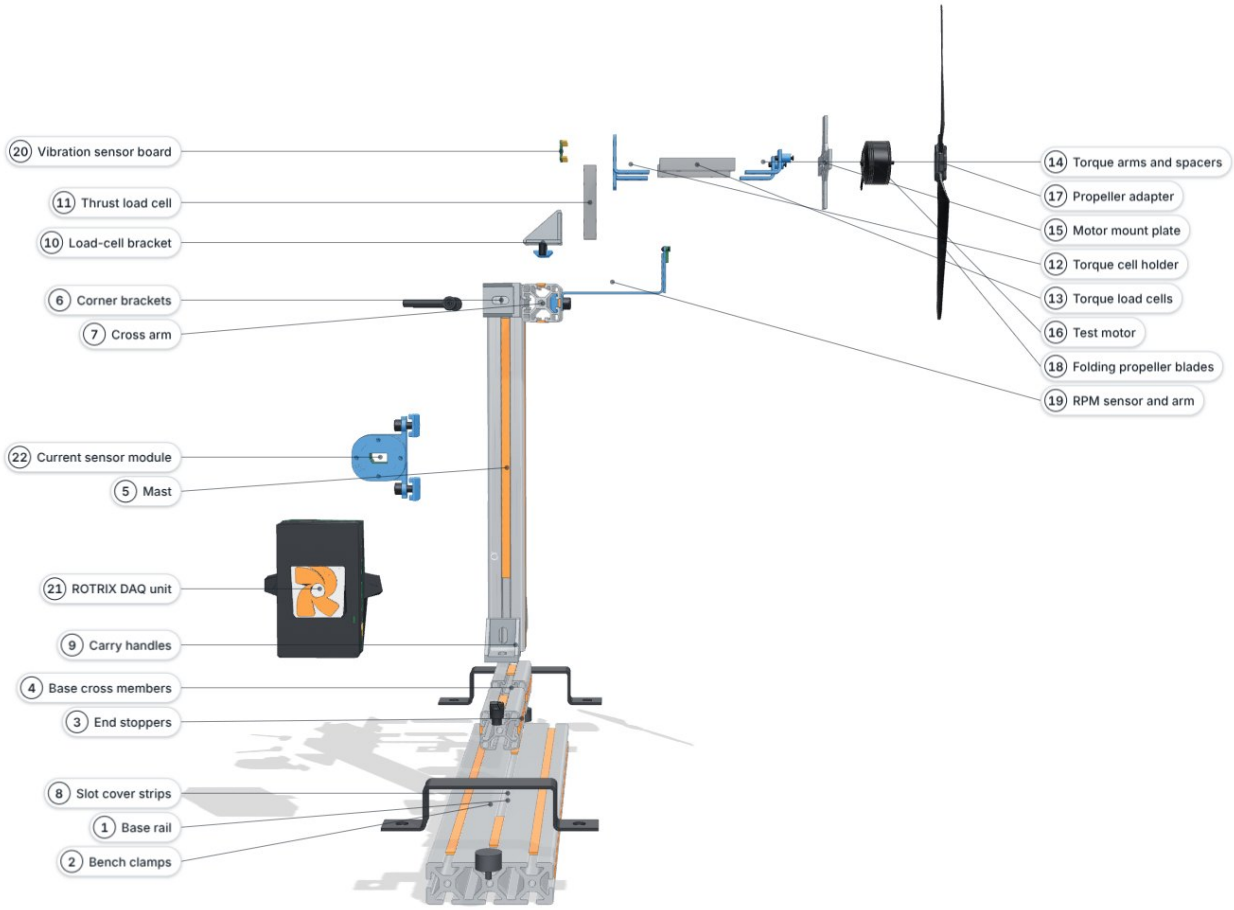


Figure 2.2 Exploded side view. Each sub-assembly moves as one block, in the order it is assembled.

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3. Safety instructions

WARNING A spinning propeller can cause severe injury. Never approach the stand while a battery is connected, and keep everyone out of the propeller plane during a test.

Assembly team

One person can assemble the stand; two people make it easier when standing the mast.

People and protection

- Read this manual completely before assembling or operating the stand.
- Wear safety glasses and gloves during assembly. Wear safety glasses and hearing protection during every test run.
- Tie back long hair and remove loose clothing, lanyards and jewellery before approaching a running motor.
- Never reach over, under or past a propeller while a battery is connected, even if the motor is stopped.

Test area

- Place the stand on a level, rigid floor or bench and anchor it with the supplied clamps before any powered test.
- Mark an exclusion zone around the stand. Nobody may stand in the plane of the propeller disc while a battery is connected.
- Keep the area clear of loose paper, tools, cable offcuts and anything that the propeller wash can lift.
- Test indoors or under cover. The sensors and the DAQ unit are not weatherproof: keep them away from rain, condensation and dust.

Electrical and battery

- Connect the battery last and disconnect it first. Keep the throttle at zero and the ESC disarmed while connecting.
- Use cables, connectors and fuses rated above the maximum test current. Check insulation before every session.
- Fit an emergency-stop or kill switch in the power line, within reach of the operator, and test it before each session.
- Charge, store and transport lithium batteries in fire-safe bags or containers, and never leave a charging battery unattended.
- When the operator works at a distance, only the digital data link and the emergency-stop loop should run to the operator station. Do not extend the analogue sensor cables.

Operation

- Never exceed the rated thrust of the stand. Stop the test if a reading approaches the limit.
- Before each run, check that every bolt is tight, the propeller is undamaged and balanced, and its hub bolts are secure.
- Start every new motor and propeller combination at low throttle and increase in steps while watching vibration and temperature.
- Stop immediately on unusual noise, rising vibration, smoke, or any reading that does not make sense.

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- Do not lean on, lift or carry the stand by the sensing head. Loads on the head can permanently damage the load cells.

CAUTION The rated thrust of this stand is 5 kgf. Loads above the rating can permanently damage the load cells and void the calibration.

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4. Package contents

Check every item against this list before you start. If anything is missing or damaged, contact REUDE support before assembly. Do not start with incomplete or faulty parts.

No.	Component	Code	Size	Material	Qty
1	**Base rail**	40 × 120 aluminium extrusion	40 × 120 mm section, 600 mm long	Anodised aluminium extrusion	1
2	**Bench clamps**	Hold-down clamps	205 × 43 × 30 mm	Steel clamp plate	2
3	**End stoppers**	Profile end caps	25 × 25 × 25 mm	Moulded rubber	2
4	**Base cross members**	40 × 40 profile	40 × 40 × 120 mm	Anodised aluminium extrusion	2
5	**Mast**	40 × 40 profile	40 × 40 × 400 mm	Anodised aluminium extrusion	1
6	**Corner brackets**	40 series 90° brackets	40 × 40 × 35 mm	Die-cast aluminium	4
7	**Cross arm**	40 × 40 profile	40 × 40 × 120 mm	Anodised aluminium extrusion	1
8	**Slot cover strips**	Orange T-slot covers · REUDE standard	9 mm wide, cut to length for each exposed slot	Flexible PVC, REUDE orange	1 set
9	**Carry handles**	Profile handles	About 57 mm long	Moulded handle with steel screws	4
10	**Load-cell bracket**	Folded sheet-metal cradle	40 × 55 mm, 3 mm sheet	Folded steel sheet	1
11	**Thrust load cell**	Beam load cell	80 × 13 × 12 mm	Precision strain-gauge load cell	1
12	**Torque cell holder**	Printed carrier	38 × 55 × 100 mm	3D-printed polymer	1
13	**Torque load cells**	Beam load cell pair	80 × 13 × 12 mm each	Precision strain-gauge load cells	2
14	**Torque arms and spacers**	Torque linkage	L-brackets 38 × 25 mm, spacers 10 mm	3D-printed polymer	4
15	**Motor mount plate**	Motor interface	100 × 100 × 3 mm	Aluminium plate	1
16	**Test motor**	MN4110 (reference)	Ø53 × 37 mm	Brushless DC motor	1
17	**Propeller adapter**	MF1302 hub	Hub about 41 mm	Aluminium hub, nylon washers, steel screws	1 set
18	**Folding propeller blades**	CW pair (reference)	About 340 mm diameter	Carbon-reinforced composite	2
19	**RPM sensor and arm**	Optical speed sensor	Arm about 112 mm	3D-printed arm, optical sensor	1
20	**Vibration sensor board**	Accelerometer	About 22 × 33 mm on 6 mm spacers	Circuit board, brass spacers	1
21	**ROTRIX DAQ unit**	STM32-based acquisition	125 × 140 × 41 mm	Custom PCB in a printed enclosure	1
22	**Current sensor module**	Inline current sensor	58 × 90 × 27 mm case	3D-printed case, current sensor board	1

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Brackets and fixings

No.	Fixing	Code	Size	Material	Qty
23	**T-nuts**	Printed, 40 series	About 11 × 21 × 25 mm	3D-printed polymer	5
24	**M8 socket cap screws**	M8 × 12 and M8 × 16	M8 × 12 / M8 × 16	Steel, black oxide	6
25	**M3 screws**	Cap and countersunk	M3 × 5 / M3 × 6	Steel	4

Component importance

Critical parts are in the load path or the measurement chain. **Important** parts set the stiffness and alignment of the frame. **Standard** parts are for convenience and protection.

No.	Component	Importance	Why it matters
1	Base rail	**Critical**	Its width and length resist the tipping force of up to 5 kgf of thrust acting high above the bench.
2	Bench clamps	**Critical**	An unclamped stand can slide or lift at full power, which ruins readings and is unsafe.
3	End stoppers	**Standard**	They keep T-nuts from sliding out and protect the bench surface.
4	Base cross members	**Important**	They give the mast a wide, stiff foot so it does not rock forward under thrust.
5	Mast	**Critical**	It carries all of the thrust down to the base. If it is not vertical the thrust line tilts and the reading is wrong.
6	Corner brackets	**Important**	They set the squareness of the mast joints, which keeps the thrust axis horizontal.
7	Cross arm	**Important**	It is the platform the whole sensing head sits on, so it must be level and rigid.
8	Slot cover strips	**Standard**	They keep dust, swarf and small parts out of the slots, cover the sharp slot edges, hold cables neatly, and give the stand its signature orange accent.
9	Carry handles	**Standard**	Makes the stand easy to carry. Handles are outside the load path.
10	Load-cell bracket	**Critical**	It is the fixed end of the thrust measurement. Any flex here reads as false thrust.
11	Thrust load cell	**Critical**	Every thrust figure the stand reports comes from this sensor.
12	Torque cell holder	**Important**	It keeps the thrust and torque sensors aligned so each reads only its own force.
13	Torque load cells	**Critical**	Reaction torque is read as a force couple across the pair. It is needed for mechanical power and efficiency.
14	Torque arms and spacers	**Important**	They make sure the torque cells see only twisting force from the motor.
15	Motor mount plate	**Critical**	It passes thrust and torque from the motor into the sensors while keeping the motor square.
16	Test motor	**Important**	This is the unit under test. Any motor within the 5 kgf rating can be fitted.
17	Propeller adapter	**Critical**	Loose blades at speed are a safety hazard, and a poorly seated hub adds vibration.

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No.	Component	Importance	Why it matters
18	Folding propeller blades	Critical	The propeller sets the load. Keep the disc clear of people and cables during every run.
19	RPM sensor and arm	Important	Motor speed is needed for efficiency figures and thrust-versus-RPM curves.
20	Vibration sensor board	Important	Picks up imbalance and resonance, which helps catch a damaged propeller or a loose motor.
21	ROTRIX DAQ unit	Critical	It samples thrust, torque, speed, voltage, current and vibration together and sends them to your PC over USB.
22	Current sensor module	Critical	Current is half of electrical power, so it is needed for every efficiency figure.

All components laid out

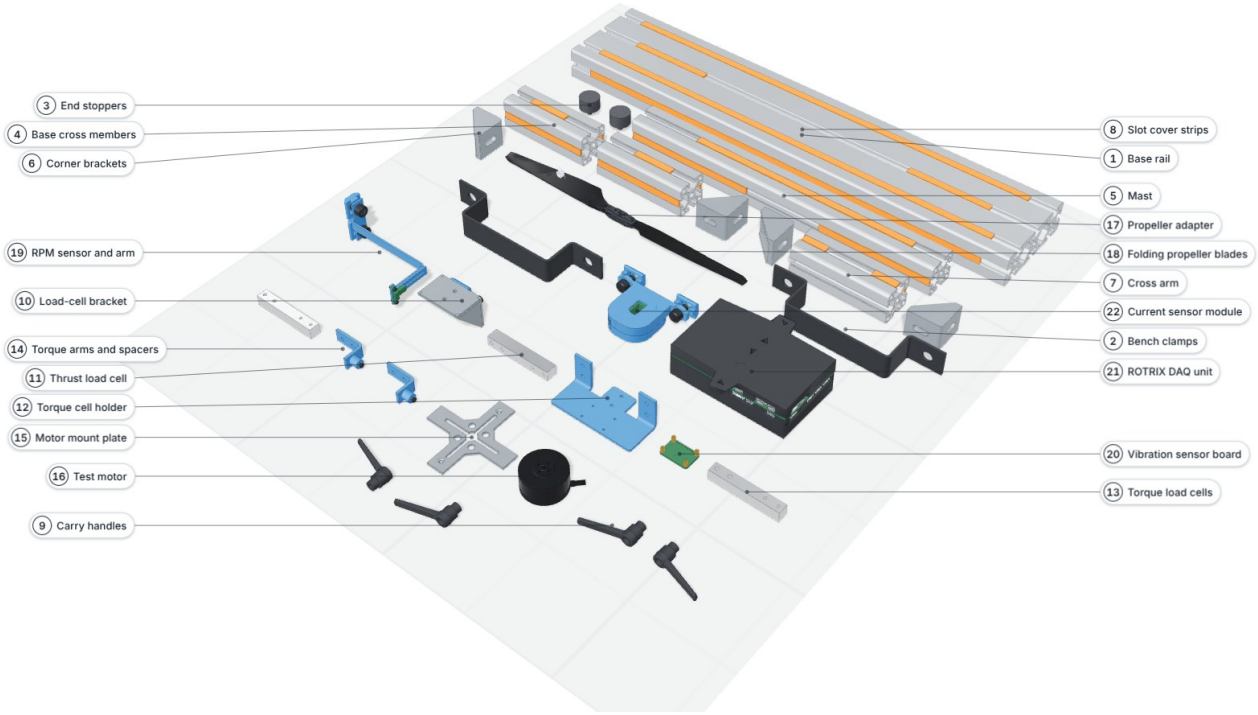


Figure 4.1 The components laid out on the table, as shipped.

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5. Tools and tightening torques

Tools you will need

- Hex keys 2.5, 4 and 5 mm
- Spanner or socket set
- Torque wrench (optional, recommended)
- Spirit level and try square
- Cable ties and a side cutter
- Multimeter
- Calibration weights (for example 0.5, 1 and 2 kg) with cord and a low-friction pulley

Recommended tightening torques

Fastener	Where used	Typical tightening
M3 screws	Printed parts, sensor boards, DAQ enclosure	Hand tight, about 0.5 to 1 Nm. Do not crack the printed parts.
M6 bolts	Motor to mounting plate, plate to bracket	About 8 Nm, with thread-locker on motor bolts
M8 button-head bolts	Brackets into T-slot nuts	About 12 to 15 Nm into steel T-nuts; snug only (about 4 Nm) into printed T-nuts. Tighten evenly around each joint.

NOTE Tighten in two passes: first snug every bolt in a joint, then bring each to the final torque. This keeps the frame square.

CAUTION Always load the T-slot nuts into a profile before a bracket, end cap or another profile closes that slot. A missing nut means taking the joint apart again.

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6. Assembly, step by step

Work through the steps in order. In each figure the parts added in that step are highlighted in orange; parts fitted in earlier steps are shown in their final position.

Step 1. Base rail, end stoppers and bench clamps

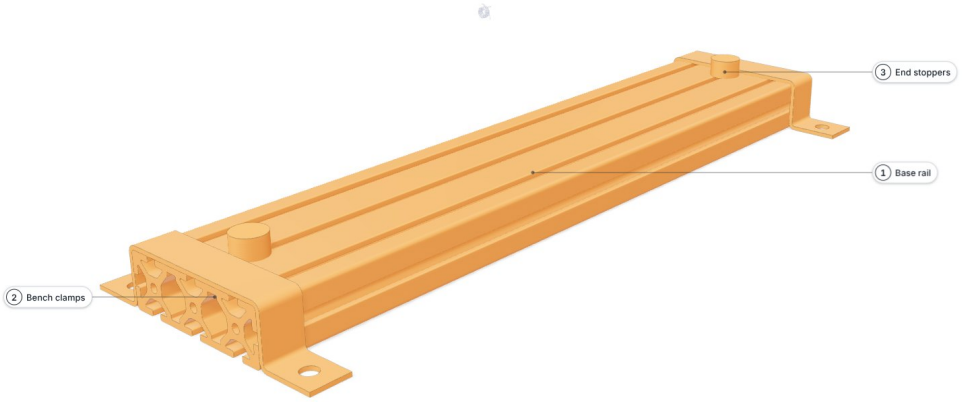


Figure 6.1 Base rail, end stoppers and bench clamps.

1. Slide T-nuts into the top slots of the base rail for the two cross members and the two base handles. Count them before closing the ends.
2. Press an end stopper into each end of the rail.
3. Place the rail on a flat, level bench with the clamps over each end.
4. Fix both clamps to the bench and tighten evenly so the rail cannot move.

CAUTION The T-nuts cannot be added once the stoppers and cross members close the slots.

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Step 2. Base cross members and base handles

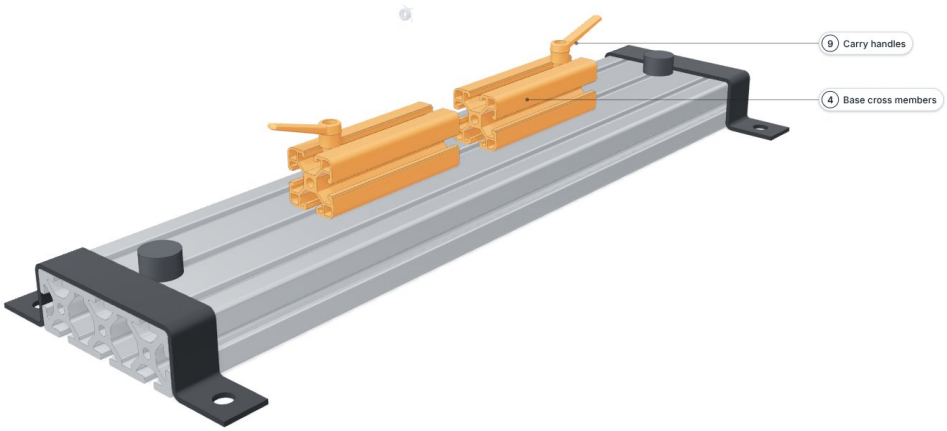


Figure 6.2 Base cross members and base handles.

1. Place the two 40 × 40 × 120 mm cross members on the base rail, one either side of where the mast will stand.
2. Bolt them into the pre-loaded T-nuts, snug but not fully tight.
3. Screw a carry handle onto the outer end of each cross member.

Step 3. Mast and lower corner brackets

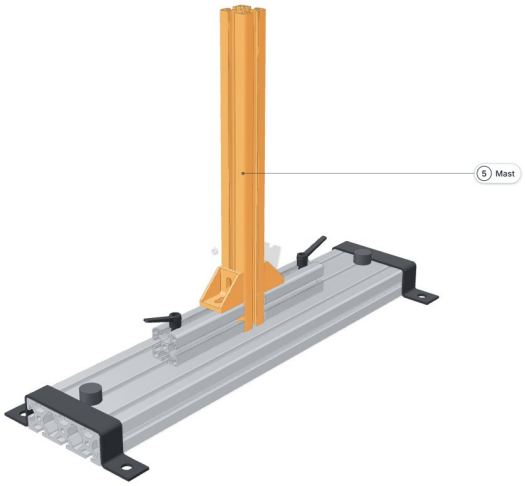


Figure 6.3 Mast and lower corner brackets.

1. Load T-nuts into the mast for the current sensor module, the DAQ unit and the cable ties.
2. Stand the 400 mm mast between the cross members.
3. Fit a corner bracket on each side of the mast foot and bolt it to the mast and the cross member.
4. Check the mast is vertical on two faces with the spirit level, then tighten all base bolts.

CAUTION Check the mast is vertical before tightening. A leaning mast tilts the thrust axis and adds error to every reading.

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Step 4. Cross arm, upper corner brackets and top handles

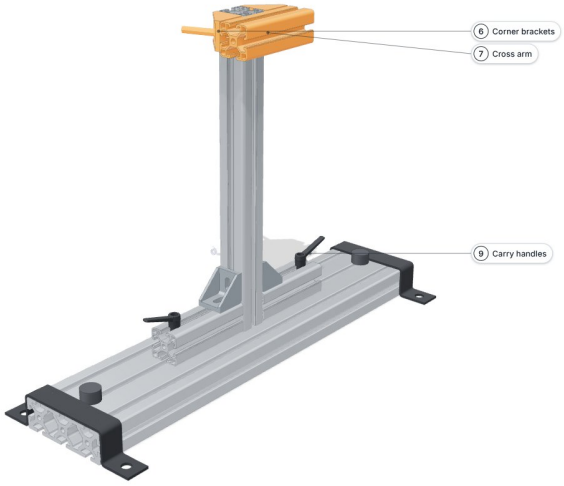


Figure 6.4 Cross arm, upper corner brackets and top handles.

1. Load T-nuts into the cross arm for the load-cell bracket and the RPM sensor arm.
2. Fix the cross arm to the top of the mast with two corner brackets.
3. Fit the two top handles.
4. Check the cross arm is level.

Step 5. Load-cell bracket, thrust load cell and torque cell holder

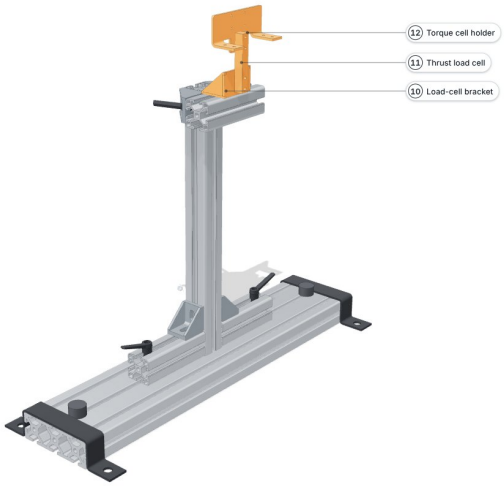


Figure 6.5 Load-cell bracket, thrust load cell and torque cell holder.

1. Bolt the load-cell bracket to the cross arm with two M8 screws.
2. Fix the thrust load cell to the bracket with its arrow pointing in the thrust direction.
3. Bolt the torque cell holder to the free end of the thrust load cell.

CAUTION Handle load cells gently. Never use them as a handle and never apply more load by hand than the rated thrust.

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Step 6. Torque load cells, torque arms and motor mount plate

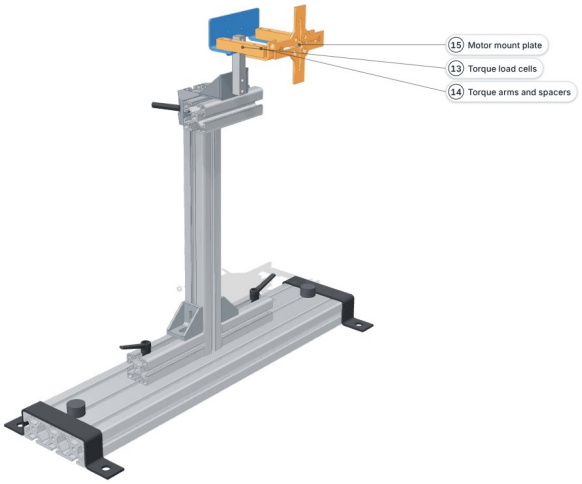


Figure 6.6 Torque load cells, torque arms and motor mount plate.

1. Fix one torque load cell to each side of the holder.
2. Fit the 10 mm spacers and the L-shaped torque arms, and secure them with the M3 screws.
3. Bolt the 100 × 100 mm motor mount plate to the torque arms.
4. Check that the plate is square to the thrust axis and touches nothing but the torque arms.

Step 7. Motor and propeller

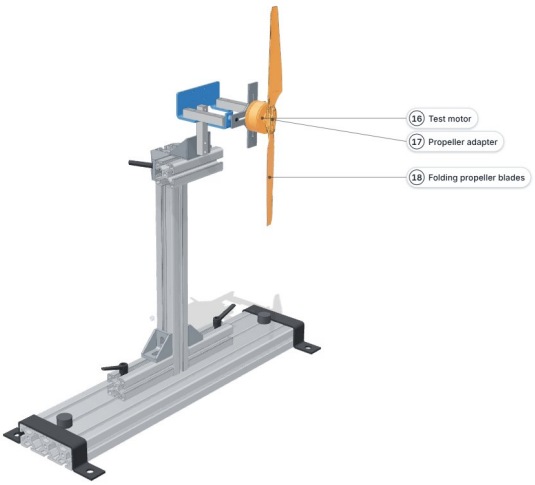


Figure 6.7 Motor and propeller.

1. Bolt the motor to the centre of the mount plate.
2. Fit the propeller adapter and blades, with a nylon washer on each side of each blade.
3. Tighten the shoulder screws evenly and check that the blades fold out freely.

CAUTION Fit the propeller last, after all wiring is complete, and only when the battery is disconnected.

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Step 8. Sensors and DAQ unit

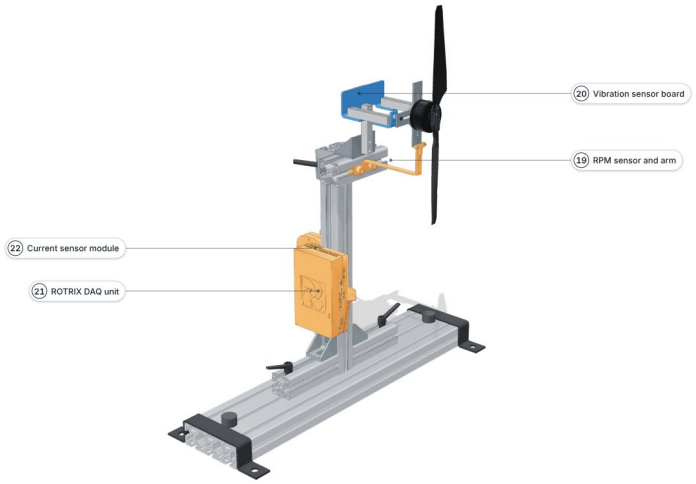


Figure 6.8 Sensors and DAQ unit.

1. Bolt the RPM sensor arm to the cross arm and aim the sensor at the rotating part of the motor or the blade path.
2. Fix the vibration board on its four spacers on the back of the torque cell holder.
3. Bolt the current sensor module to the rear face of the mast.
4. Screw the ROTRIX DAQ unit to the side face of the mast, below the cross arm, with its connectors facing up.

CAUTION The DAQ unit is supplied assembled and sealed. Do not open the enclosure.

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Step 9. Orange slot cover strips (REUDE standard)



Figure 6.9 Orange slot cover strips (REUDE standard).

1. Check that every bolt in the frame is at its final torque and that all cables are routed.
2. Press an orange cover strip into each open T-slot, starting at one end and working along by hand.
3. Cut the strip flush at each bracket and at the profile ends with a sharp knife.
4. Where a cable runs inside a slot, lay it in the slot first and press the cover over it.

CAUTION Fit the covers last. They are easy to lift out later if you need to add a T-nut or re-route a cable.

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Sensing head, exploded

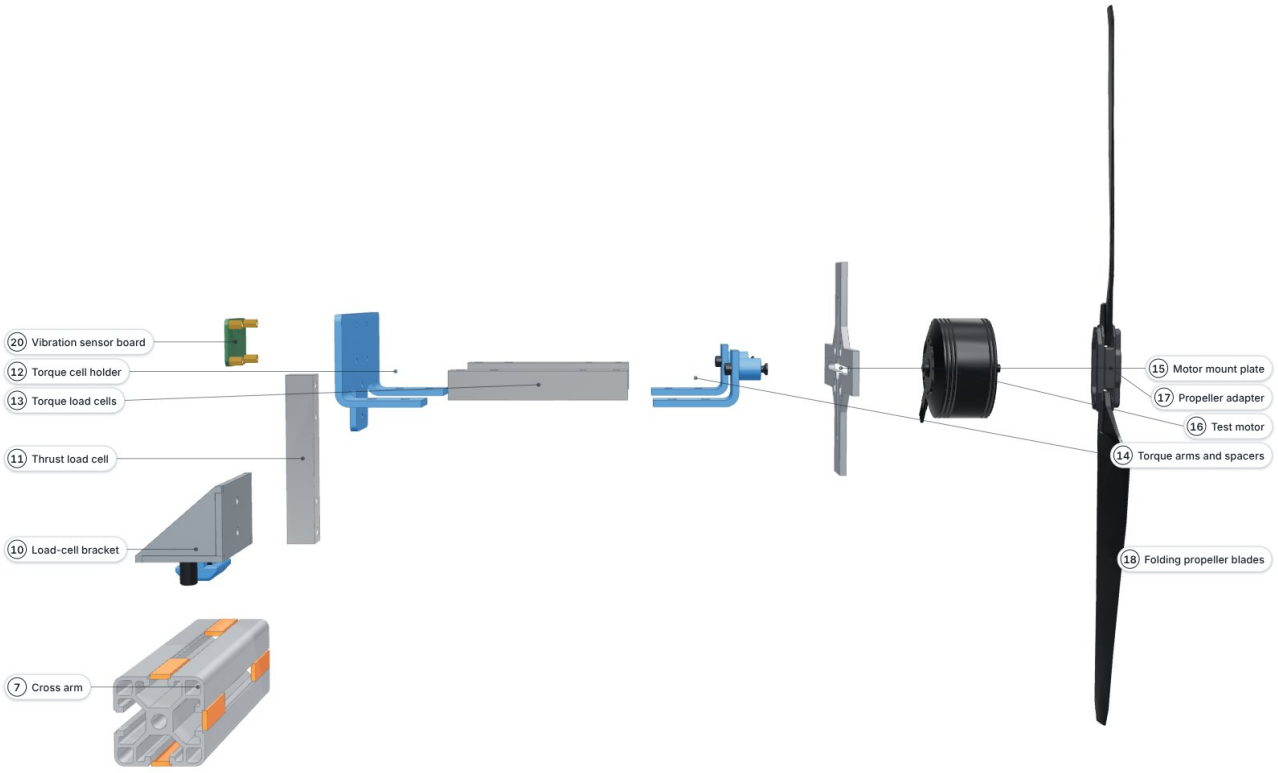


Figure 6.10 Exploded side view of the force-sensing head, motor and propeller.

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7. DAQ and electronics assembly

The Compact 5 carries the ROTRIX DAQ unit, the same unit as on the RTS Pro 20, on the side face of the mast below the cross arm. Every sensor cable runs down the rear face of the mast and round the corner into the top of the DAQ unit, so the cables stay short.

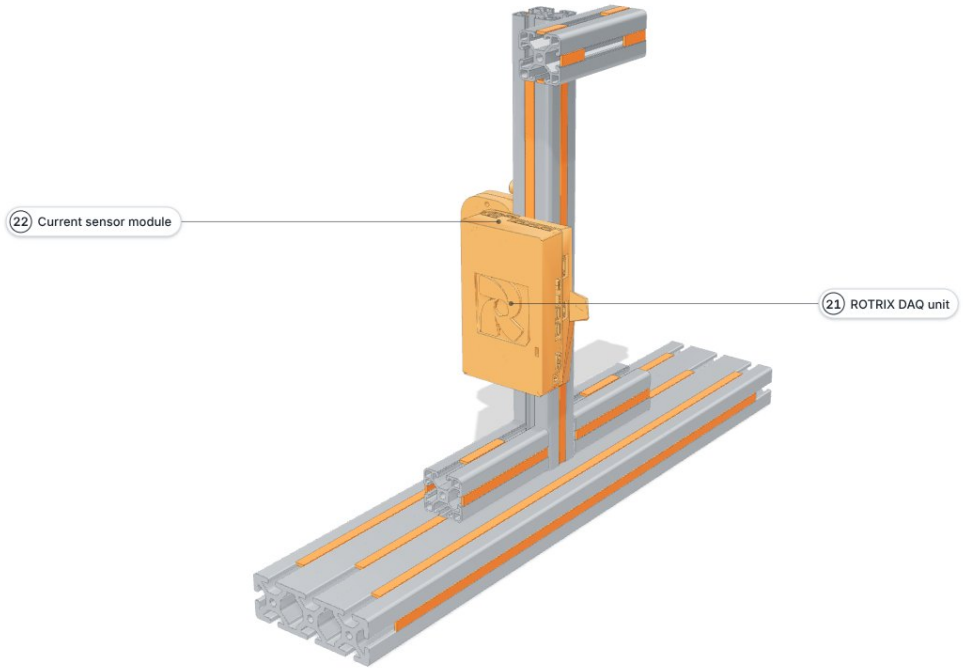


Figure 7.1 ROTRIX DAQ unit on the side of the mast, with the current sensor module on the rear face.

Mounting and connecting

1. Slide two T-nuts into the side slot of the mast and screw the DAQ unit on, about 60 mm below the cross arm, connectors facing up.
2. Fit the current sensor board in its case on the mast, close the cover and screw it shut.
3. Run the battery-to-ESC lead through the current sensor, with the arrow pointing from the battery to the ESC.
4. Connect the thrust and torque load cells, RPM sensor, vibration board and current sensor to the matching inputs on the DAQ unit.
5. Connect the DAQ unit to the PC with the USB cable.

What the DAQ unit measures

- Thrust and torque from the load cells, through precision bridge amplifiers.
- Motor speed from the optical RPM sensor.
- Battery voltage and current in the power line to the ESC.
- Vibration and temperature from the sensor boards.
- All channels are time-stamped together and streamed to ROTRIX for live display, logging and reports.

CAUTION Connect and disconnect sensor cables only with the DAQ unit powered off and the battery disconnected.

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8. Wire routing to the sensors

Good cable routing matters as much as good mechanics. A cable that is pulled tight across the sensing head carries part of the load, and the load cells then measure the cable instead of the motor. Follow the routes and rules below.



Figure 8.1 Recommended cable routes. Signal cables and power leads run on separate sides of the frame.

	Cable
	Thrust load cell
	Torque load cells
	RPM sensor
	Vibration sensor
	Current sensor
	Battery power (through the current sensor)
	USB to PC

Routing table

Cable	From	Route	To
Thrust and torque load cells	Sensing head	Loose service loop at the head, down the rear face of the mast, round the corner into the top of the DAQ unit	DAQ load-cell inputs
RPM sensor	RPM arm	Along the cross arm, down the rear face of the mast	DAQ RPM input

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Cable	From	Route	To
Vibration sensor	Back of the torque cell holder	Service loop, then down the mast with the load-cell cables	DAQ vibration input
Current sensor signal	Current sensor module	Short lead round the mast corner to the DAQ unit	DAQ current input
Power leads	Battery	Through the current sensor, up the rear of the mast and over the cross arm, kept apart from the signal cables	ESC, then motor
USB	DAQ unit	From the bottom of the DAQ unit, along the base rail and off the bench	PC

Routing rules

1. **Leave a service loop.** Where a cable leaves the sensing head, form a loose loop about 50 mm across, so the cable carries no load when the head deflects.
2. **Tie to the fixed frame only.** Fix cables to the frame with ties about every 150 mm. Never tie a cable to the motor plate, the load cells or the free end of the sensing head.
3. **Keep signal and power apart.** Run sensor cables and power leads on opposite sides of the frame. Where they must cross, cross at a right angle.
4. **Stay out of the propeller disc.** Route every cable behind the motor plate and away from the plane of the propeller.
5. **Relieve strain at the DAQ.** Leave a little slack at each connector and label both ends of every cable.
6. **Screened load-cell cables.** Connect the screen at the DAQ end only.
7. **Remote operation.** If the operator sits away from the stand, extend only the USB data link and the emergency-stop loop. Never extend the analogue sensor cables.
8. **Current sensor direction.** Fit the current sensor with its arrow pointing from the battery towards the ESC.
9. **Use the slots.** On the fixed frame, lay cables inside the T-slots and press the orange slot covers over them. This keeps cables tidy, protected and clear of the propeller.

CAUTION After any change to the wiring, zero the load cells again and check that the zero reading stays steady for one minute before testing.

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9. Commissioning and calibration

Before first power-up

- All bolts tightened to the recommended torque, and the base anchored.
- Motor fitted and centred; propeller not yet fitted.
- All cables routed with service loops, and the emergency stop tested.

Connect and check the sensors

1. Connect the DAQ unit to the PC over USB and open ROTRIX.
2. Confirm that every sensor is detected and shows a sensible live value.
3. Allow 10 minutes for the load cells to warm up, then zero (tare) thrust and torque with the motor fitted and stationary.

Check thrust and torque with known loads

1. Thrust: tie a cord to the motor hub, run it over a low-friction pulley in line with the thrust axis, and hang known weights (0.5, 1 and 2 kg). Compare each reading with the weight.
2. Torque: fit a light lever arm of known length to the motor plate and hang a known weight from its end. The expected torque is weight × 9.81 × arm length (N·m, with the weight in kg and the length in m).
3. If any reading is outside the stated accuracy, run the calibration routine in ROTRIX and repeat the check.

Check RPM, voltage and current

- Compare the RPM reading with a hand tachometer at two or three throttle settings.
- Compare voltage and current with a multimeter and a clamp meter.

First run

1. Fit and balance the propeller, and clear the exclusion zone.
2. Connect the battery last. Arm the ESC from the operator station.
3. Run at about 10% throttle for 30 seconds, watching vibration, temperature and noise.
4. Increase in steps of 10% only while every reading stays stable.

10. Operating the stand

- Do not exceed 5 kgf of thrust. Set a thrust limit alarm in ROTRIX below the rating.
- Run a short warm-up before each test series and zero the load cells with the motor stopped.
- Use ROTRIX to run automated throttle steps and ramps, and ROTRIDASH to compare runs and produce PDF performance reports.
- Allow the motor and ESC to cool between long runs, and watch the temperature channels.
- After testing, disarm the ESC, disconnect the battery, then stop logging and save the data.

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11. Maintenance

When	Check
Before every session	Bolts on the sensing head and motor; propeller condition and balance; cable routing and service loops; clamps anchored; emergency stop works
Weekly in regular use	All frame bolts at the recommended torque; sensors clean; no cracks in brackets or printed parts
Every month	Zero and span check of the thrust and torque readings with known weights; RPM check against a hand tachometer
Every 6 months	Full calibration in ROTRIX; inspect cables and connectors; replace worn fasteners
After a crash or overload	Stop testing. Inspect the load cells and brackets, then run a full calibration before further use

NOTE Clean the sensors and the motor plate with a dry cloth after each session. Do not use solvents on the load cells or the circuit boards.

12. Troubleshooting

Symptom	Likely cause	What to do
Thrust reading drifts with the motor stopped	Load-cell cable under tension, or the load cells are still warming up	Re-route the cable with a loose service loop, wait 10 minutes after power-up, then zero again
Readings are noisy or jumpy	Loose bolts, an unbalanced propeller, or signal cables running beside power leads	Retighten the frame, balance the propeller, separate signal cables from the power leads
Torque reads a value with no load	Sensor block not square to the thrust axis, or the motor plate is touching the frame	Check alignment and clearances, then zero again
RPM shows zero or jumps	Sensor not aimed at the blade path, or too far away	Re-aim the sensor and move it closer, keeping it clear of the blade tips
DAQ unit not found by ROTRIX	USB cable, port or driver	Try another cable and port, check the driver, and restart ROTRIX
The stand creeps or rocks during a run	Clamps not anchored, or the floor is uneven	Stop, level the base and anchor every clamp

13. Support and contact

Contact	Details
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When contacting support, quote the document ID on the cover, the stand serial number, and the ROTRIX software version.

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